Regression Project

Duration	Team Size	Language	Deliverables
2 Weeks	**10 Members**	**Python Only**	**3 (Code, Report)**

1. Overview

You are tasked to identify a messy, imperfect dataset, perform rigorous preprocessing, and then build a complete regression pipeline including model fitting, comparison, and **interactive** visualizations. The core requirement is analytical rigor: your preprocessing decisions must be deliberate and documented, and your regression results must be explainable in the context of the data you chose.

2. System Requirements

2.1 Dataset Selection & Justification

What Makes a Dataset Suitable:

- Must have a clear numeric target variable (Y) suitable for regression.
- Must contain at least 5 numeric/encodable predictor variables
- Must have at least 200 rows of data
- Must be sourced from a reputable public source.
- Must be intentionally messy, containing at least three of the following:
  - Missing values in one or more columns
  - Categorical columns that require encoding
  - Outliers or anomalous entries
  - Redundant or irrelevant columns
  - Mixed or inconsistent data formats

2.2 Data Preprocessing

Here you will create all necessary preprocessing steps to prepare your data for analysis and modelling. All preprocessing decisions must be justified in your report.

2.3 Model Fitting (Implemented from Scratch)

Both methods below must be implemented manually using NumPy. Use of sklearn, statsmodels, or any regression library for the core math is not permitted.

- **Numerical View**
- **Gradient Descent**
- **Statistical View**

2.4 Model Comparison

Given the preprocessed dataset and the target variable Y, your code should generate and compare at least 3 model specifications, each model specification should be tested with both regularization methods (Lasso and Ridge):

1. Best single predictor (the feature with the highest absolute correlation with Y).
2. Full model (all available numeric and encoded categorical predictors).
3. A domain-knowledge based selection.

Results must be reported side-by-side in a comparison table showing R^2, Adjusted R^2, and number of predictors.

2.5 Visualization

All visualizations should be interactive (Plotly recommended) and rendered programmatically:

- Correlation heatmap across all numeric variables
- Scatter plot: user-selectable X vs Y with fitted regression line
- Gradient Descent convergence plot: loss vs. iterations
- Residual plot for each of the 3 model specifications
- Coefficient bar chart comparing betas across model specs

3. Deliverables

Submit a single Jupyter Notebook (.ipynb) containing all code, outputs, and markdown explanations, alongside a written report (digital format not printed) that covers the full project: dataset justification, preprocessing decisions, mathematical derivation of both models, comparison results, and a discussion of what the regression reveals about the chosen data.

4. Rules & Constraints

- Python only. No other languages.
- All regression implementations must from scratch using NumPy. **Do not use** sklearn, statsmodels, or any regression library for the core math. Visualization and data loading libraries are permitted.
- All team members must contribute meaningfully to the notebook.